

CSE 4631: Digital Signal Processing

Class Test 3 & 4

Total Marks: 40

Instructions:

1. Answer all questions.
2. Show necessary mathematical steps for numerical problems.
3. Assume all signals are discrete-time unless otherwise stated.
4. Draw relevant diagrams wherever necessary.
5. Clearly mention all assumptions made during the solution.

Question 1: Discrete-Time Signals and Systems

[8 Marks]

(a) Given the discrete-time signal

$$x[n] = \{2, -1, 3, 4, -2\}$$

where the sample $x[0] = 3$, determine and sketch the following signals:

- i. $x[n]$
- ii. $x[n - 2]$
- iii. $x[-n]$
- iv. $x[2 - n]$

[5 Marks]

(b) Determine whether the following system is **linear, time-invariant, causal, and stable**:

$$y[n] = nx[n] + x[n - 1]$$

Justify your answers briefly.

[3 Marks]

Question 2: Linear Convolution

[8 Marks]

Consider the two finite-duration signals:

$$x[n] = \{1, 2, 1\}$$

and

$$h[n] = \{1, -1, 2\}$$

where the first element of each sequence corresponds to $n = 0$.

- (a) Calculate the output

$$y[n] = x[n] * h[n]$$

using the **linear convolution** method.

[5 Marks]

- (b) Verify your result using the tabular or graphical convolution method.

[3 Marks]

Question 3: Difference Equation and LTI System

[8 Marks]

An LTI discrete-time system is described by

$$y[n] - 0.5y[n-1] = x[n] + x[n-1]$$

with initial rest conditions.

- (a) Determine the impulse response $h[n]$ of the system.

[4 Marks]

- (b) If the input is

$$x[n] = u[n],$$

determine the output $y[n]$.

[4 Marks]

Question 4: Sampling and Aliasing

[8 Marks]

A continuous-time signal is given by

$$x(t) = \cos(2\pi(2 \text{ kHz})t) + 0.5 \cos(2\pi(5 \text{ kHz})t).$$

- (a) Determine the minimum sampling frequency required to avoid aliasing.

[2 Marks]

- (b) If the signal is sampled at

$$f_s = 6 \text{ kHz},$$

determine the apparent frequency of each sinusoidal component after sampling.

[4 Marks]

- (c) Explain briefly what **aliasing** is and how an anti-aliasing filter can prevent it.

[2 Marks]

Question 5: Z-Transform and System Analysis

[8 Marks]

Consider the discrete-time signal

$$x[n] = (0.5)^n u[n].$$

- (a) Find the **Z-transform** $X(z)$ and specify its region of convergence (ROC).

[3 Marks]

(b) Determine the inverse Z-transform of

$$X(z) = \frac{z}{z - 0.8}$$

for the ROC

$$|z| > 0.8.$$

[2 Marks]

(c) For the system

$$H(z) = \frac{1}{1 - 0.6z^{-1}},$$

determine whether the system is stable for the following ROC:

$$|z| > 0.6.$$

Give a reason for your answer.

[3 Marks]

Mark Distribution

Question	Topic	Marks
Q1	Discrete-Time Signals and LTI Systems	8
Q2	Linear Convolution	8
Q3	Difference Equation and Impulse Response	8
Q4	Sampling and Aliasing	8
Q5	Z-Transform and System Analysis	8
Total		40

Topics Covered:

- Discrete-time signal manipulation
- Classification of discrete-time systems
- Linearity and time invariance
- Causality and stability
- Linear convolution
- LTI systems
- Difference equations
- Impulse response
- Sampling theorem
- Aliasing
- Z-transform
- Region of convergence (ROC)
- Inverse Z-transform
- Stability analysis